

CM12002

Computer Systems

Architectures

Boolean Logic and Circuits

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Boolean Logic and Circuits

In order to Simplify Boolean Formulas, we first need to cover:

- Standard Sum of Products (SofP)

Standard forms: Standard Sum of Products (SofP)

- A **standard Sum of Products** of a Boolean formula is:
 - a series of terms joined by OR operations, in which
 - each term contains a combination of all variables joined by AND operations

$$\text{e.g.: } A + B.C = A.B.C + A.B.C' + A.B'.C + A.B'.C' + A'.B.C$$

Theorem:

- Every Boolean expression over n variables is equivalent to a *standard sum of products*.
- **n.b.** Here, equivalent means both “equivalent with respect to truth tables” **and** “equivalent with respect to algebraic manipulation”.

Let's see how we arrived at the above Sum of Products

Boolean algebra

Boolean equalities

Name	AND rule	OR rule	NOT rule
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$A.0 = 0$	$A+0 = A$	$0' = 1$
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$A.1 = A$	$A+1 = 1$	$1' = 0$
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$A.A = A$	$A+A = A$	$(A')' = A$
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$A.A' = 0$	$A+A' = 1$	
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Commutative	$A.B = B.A$	$A+B = B+A$
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Associative	$A.(B.C) = (A.B).C$	$A+(B+C) = (A+B)+C$
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Distributive	$A.(B+C) = A.B+A.C$	$A+(B.C) = (A+B).(A+C)$
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De Morgan	$(A.B)' = A'+B'$	$(A+B)' = A'.B'$
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Standard Sum of Products

An Example

$$A + B.C$$

$$= A.(B + B') + B.C$$

$$= A.B + A.B' + B.C$$

$$= A.B.(C+C') + A.B' + B.C$$

$$= A.B.C + A.B.C' + A.B' + B.C$$

$$= A.B.C + A.B.C' + A.B'.(C+C') + B.C$$

$$= A.B.C + A.B.C' + A.B'.C + A.B'.C' + B.C$$

$$= A.B.C + A.B.C' + A.B'.C + A.B'.C' + B.C.(A+A')$$

$$= A.B.C + A.B.C' + A.B'.C + A.B'.C' + A.B.C + A'.B.C$$

$$= A.B.C + A.B.C' + A.B'.C + A.B'.C' + A'.B.C$$

$$A + A' = 1$$

Which is a standard sum of products:

- a series of terms joined by OR operations, in which
- each term contains a combination of all variables joined by AND operations

Standard Sum of Products

By truth table:

- for each row of the truth table with value 1, form the product of all literals with value 1, then form the sum of all these products.

our example: $A + B.C$

A	B	C	(B.C)	A+(B.C)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$(A'.B.C) + (A.B'.C') + (A.B'.C) + (A.B.C') + (A.B.C)$$

Standard Sum of Products

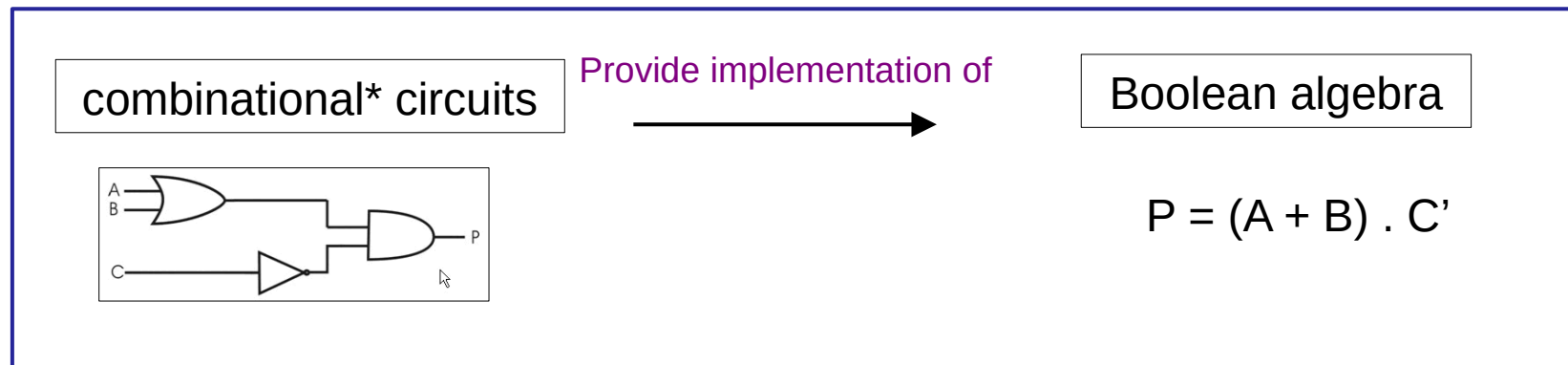
From $A + B.C$ to $A.B.C + A.B.C' + A.B'.C + A.B'.C' + A'.B.C$

Why make things more complicated?

Simplifying Boolean Formulas

Boolean logic

Recap: Modern *logic gates* are electronic switches which correspond to the *operators* of Boolean algebra.



Simplification

- To implement a formula of Boolean logic **efficiently** as a circuit
 - we need to put it in its simplest form.
- One approach
 - algebraic manipulation using the equations of Boolean algebra

Two problems:

- How do we know which rules to apply?
- How do we know when we have reached a simplest form?
 - E.g., how do we show that $A + B.C$ is indeed the simplest form of this original Boolean function $(A+B+C).(A+B')$?

Simplification

We will develop simplification as a two stage process :

1. First convert the Boolean formula to a ***standard sum of products***.
2. Then find the simplest form for this formula (the one with the fewest logic gates).